

Report – Project Description

This project involved creating a program in C++ that searches a database of images to find visually similar ones based on a target image. Using the OpenCV library and various feature extraction methods, the program compares images using different techniques ranging from simple pixel comparison to AI based feature vectors. The program takes a target image, a database of images, a number of matches to return, and a matching method as inputs and returns the most similar images in order of similarity.

Various matching methods were implemented including a baseline method that compares a small center patch of two images, a color histogram method that compares overall color distribution, a multi histogram method that compares color from different regions of the image, and a texture and color method that combines color and edge patterns. Additionally a deep neural network embedding method was implemented using pre computed feature vectors from a ResNet18 network. A custom method was also designed to detect images with similar structural patterns like stairs using horizontal edge density.

Required Images

Task 1



Target on left (pic.1016), Top 3 matches below (pic.0986, pic.0641, pic.0547)

Results matched professor's. Seems to work as intended as it recognizes the pattern of red in the middle.



Task 2



Target on left (pic.0164), Top 3 matches below (pic.0110, pic.1032, pic.0092)

Results seem to make sense considering a common pattern of grey buildings surrounded by blue sky.



Task 3



Target on left (pic.0274), Top 3 matches below (pic.0273, pic.0209, pic.0881)

Results mostly make sense as top 2 results showed same object which seems to be a water tank surrounded by trees and blue sky, but the 3rd matching result shows a blue handicap sign surrounded by plants.



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Task 4



Target on left (pic.0535), Top 3 matches below (pic.0171, pic.0629, pic.0853)

Results don't seem to relate to each other as it's a mix of indoors with tables and objects vs. outdoors with plants.



Task 5



Target on left (pic.0893), Top 3 matches below (pic.0897, pic.0136, pic.0885)

This is actually one of my favorites as it seems to pickup the fire hydrant photos as one would expect flawlessly.



Target on left (pic.0164), Top 3 matches below (pic.0213, pic.1032, pic.0543)

Shows a pattern for first 2 matches with grey buildings surrounded by blue sky and while the 3rd image doesn't seem as close of a match with different colors, it does still show a building with a sky overhead.



Task 6

The DNN consistently showed better results than the histogram ones where any objects from target images such as flowers, mascot doll, or construction vehicle, were more consistently picked up in the resulting top matched photos.

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Target on left (pic.0104), Top 5 matches below (pic.0233, pic.0727, pic.0818, pic.0185, pic.1016)

This custom feature converts each image to greyscale and computes horizontal edge strength per row using a Sobel Y filter with SSD distance matching, however results were inconsistent as the feature could not reliably distinguish stairs from other structured images with similar horizontal edge patterns. The thought was to use the parallel lines from sobel and try to recognize them as patterns of stairs.



Reflection

This project introduced content-based image retrieval and demonstrated how different feature extraction methods produce very different results for the same query image. The most valuable insight was seeing how deep network embeddings consistently outperformed classic histogram methods especially when searching for specific objects. While histograms worked broadly on general backgrounds and color patterns the DNN embeddings were able to capture higher level content similarity making them far more effective at finding semantically related images.

The most challenging part was implementing the custom stairs detection feature in Task 7. The idea of using horizontal edge patterns from a Sobel Y filter to detect recurring parallel lines of stairs was sound in theory but proved difficult in practice as many other structured images produced similar edge patterns making reliable stair detection inconsistent.

Overall, this project deepened my understanding of how computers can be taught to measure visual similarity and how the choice of features and distance dramatically affects the quality of results.

Acknowledgements

The following materials and resources were consulted for this project:

- Professor Bruce A. Maxwell provided sample code including `csv_util.cpp`, `csv_util.h`, and the `ResNet18_olym.csv` embeddings file.
- OpenCV documentation was referenced for Sobel filter functions and general library usage.